

SOC Internship Day 32 – Cloud Threat Detection Notes

Objective

Learn how SOC analysts detect threats in cloud environments such as AWS and Azure.

1. What is Cloud Threat Detection?

Cloud Threat Detection is the process of monitoring cloud infrastructure, user activities, and logs to identify suspicious or malicious behavior in cloud environments.

2. Importance of Cloud Threat Detection

- Protects cloud resources from attacks
- Detects unauthorized access attempts
- Helps prevent data breaches
- Improves visibility into cloud activities and security events

3. AWS Attack Monitoring

SOC analysts monitor AWS environments for:

- Unauthorized AWS Console logins
- EC2 instance compromise
- Suspicious access to S3 buckets
- Privilege escalation attempts

4. IAM Abuse Detection

IAM (Identity and Access Management) abuse occurs when attackers misuse permissions or user accounts.

Common Indicators:

- Creation of new administrator accounts
- Changes to permissions or policies
- Multi-Factor Authentication (MFA) being disabled
- Logins from unusual geographic locations

5. Suspicious API Calls

API activity can reveal attacker actions.

Examples of High-Risk API Calls:

- CreateAccessKey
- DeleteTrail
- CreateUser
- StopLogging

6. CloudTrail Analysis

AWS CloudTrail logs provide visibility into:

- API calls
- User activities
- Authentication events
- Resource modifications

These logs are essential for investigations and threat detection.

7. Azure Security Monitoring

Key monitoring areas in Azure:

- Azure AD sign-in events
- Conditional Access alerts
- Privileged Identity Management (PIM) monitoring
- Detection of abnormal user behavior

8. Cloud Detection Techniques

SOC teams use several techniques, including:

- Monitoring impossible-travel logins
- Detecting privilege escalation attempts
- Analyzing API activity
- Tracking excessive data downloads

9. Cloud SOC Workflow

Cloud Logs

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SIEM Correlation

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Threat Detection

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SOC Investigation 10. SOC Roles in Cloud Security

- **L1 Analyst:** Monitor alerts and perform initial triage
- **L2 Analyst:** Investigate suspicious activities and incidents
- **L3 Analyst:** Conduct threat hunting and incident response

Assignment Answers

1. Explain Cloud Threat Detection

Cloud Threat Detection is the process of monitoring cloud services, user activities, and logs to identify unauthorized access, malicious behavior, and security threats. It helps organizations protect cloud resources, prevent data breaches, and maintain visibility into cloud environments.

2. Research IAM Abuse

IAM abuse occurs when attackers misuse identities, roles, or permissions to gain unauthorized access. Common indicators include new admin account creation, permission changes, MFA removal, and unusual login locations. Organizations can reduce IAM abuse by enforcing least privilege, MFA, and continuous monitoring.

3. Analyze Suspicious API Calls

Suspicious API calls such as `CreateAccessKey`, `DeleteTrail`, `CreateUser`, and `StopLogging` may indicate attacker activity. Monitoring these actions helps SOC analysts detect privilege escalation, credential abuse, log tampering, and unauthorized account creation before significant damage occurs.